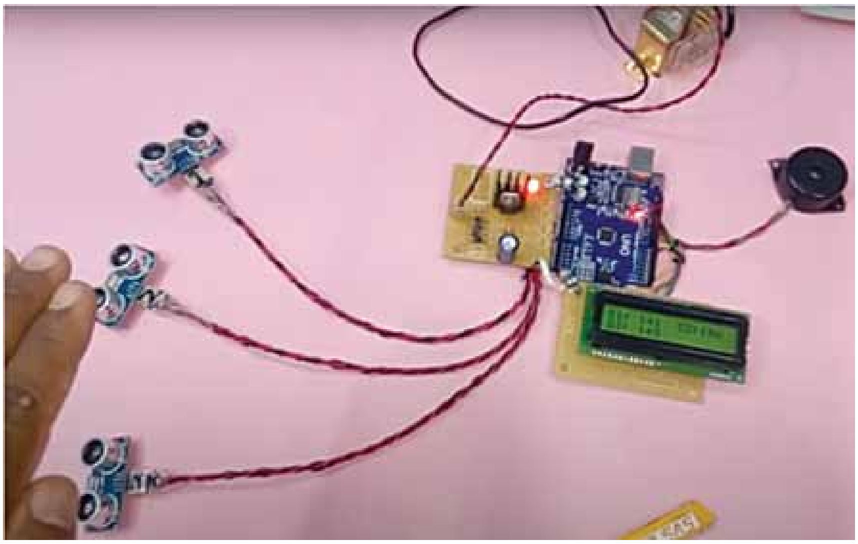

Fig. 4: Author's prototype with LCD

Construction and testing

Assemble the circuit and attach the sensors properly on the jacket. But before doing so upload the source code to Arduino using Arduino IDE.

Place ultrasonic sensor US1 on right arm of the jacket, sensor US2 on left arm, and sensor US3 and Arduino board on back side of the jacket. Connect power supply pins and input/output pins of the sensors to the Arduino board. To power up the board, a 9V PP3 battery with a 2.1mm DC plug can be used. The serial monitor utility of Arduino is used to display the measurements of all three sensors.

For testing purpose, the safe distance was kept as 30cm. You can increase the safe distance in the code as per desired physical distancing norms.

The project can be extended for use with an LCD to display the distance as shown in Fig. 4. EFY

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno
US1-US3	- HC-SR04 ultrasonic sensor module
Miscellaneous:	
BATT.1	- 9V PP3 battery
PZ1	- Piezo buzzer
	- 2.1mm DC plug
	- Jumper wires
	- Jacket

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